



SERVICE RELIABILITY MONITORING & ANALYSIS

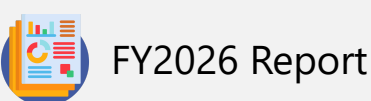
This project analyzes the Chicago Transit Authority's (CTA) bus and train service reliability using live and historical alert data. It provides a real time view of current service status across all routes, along with a persistence analysis that flags routes stuck in an unchanged status for extended periods. An 8 day historical view is included to uncover patterns in delays, disruptions, and affected routes over time broken down separately for bus and train service to highlight routes and time of day trends. The project concludes with a set of data driven recommendations, connecting these findings to CTA's own stated goals and financial constraints as outlined in its FY2026 Budget Book.

Content

- 1 Status (Live)
- 2 Persistence (Live)
- 3 Overview (Historical)
- 4 Performance (Historical)
- 5 Recommendations & Insights
- 6 Data Architecture
- 7 ERD
- 8 Project Workflow
- 9 Analytical Workflow
- 10 Data Dictionary

About

- 1 Tools: SQL Server | Power BI | SQL Server Agent | Jira | Python | VS Code
- 2 Last Updated: Sept 2026
- 3 Designed by: Ayesha Batool



Delay Rate

1.8% ▼

Share of tracked services currently reporting a delay

Services Tracked

138

Disrupted Alerts

56

Normal Services

82

Tracking 138 routes (127 Bus · 8 Rail · 3 Systemwide)

Train Lines Alert Summary

Route	Severity Level		Current Status
Green Line		Low	Special Note
Blue Line		Medium	Planned Work w/Part Closure
Brown Line		No Delay	Normal Service
Orange Line		No Delay	Normal Service
Pink Line		No Delay	Normal Service
Red Line		No Delay	Normal Service
Purple Line		Low	Added Service
Yellow Line		Low	Added Service

Systemwide Alert Summary

Route	Severity Level	Current Status
All Bus Routes	Medium	Service Change
All Routes	No Delay	Normal Service
All Train Routes	No Delay	Normal Service

Route Alert Summary

Train

Bus

Systemwide

Rail

2

4

1

1

Bus Lines Alert Summary

Route	Severity Level	Current Status
Montrose	Medium	Service Change
67th-69th-71st	Medium	Planned Reroute
Archer	Medium	Planned Reroute
Armitage	Medium	Planned Reroute
Austin	Medium	Planned Reroute
Blue Island/26th	Medium	Planned Reroute
Broadway	Medium	Planned Reroute
California/Dodge	Medium	Planned Reroute
Chicago	Medium	Planned Reroute
Clarendon/Michigan Express	Medium	Planned Reroute
Cottage Grove	Medium	Planned Reroute
Division	Medium	Planned Reroute
Foster	Medium	Planned Reroute
Halsted	Medium	Planned Reroute
Harrison	Medium	Planned Reroute
Hyde Park Express	Medium	Planned Reroute
Inner Lake Shore/Michigan Express	Medium	Planned Reroute
Jackson	Medium	Planned Reroute
Jackson Park Express	Medium	Planned Reroute
Jeffery Jump	Medium	Planned Reroute
Jeffery Local	Medium	Planned Reroute
Kimball-Homan	Medium	Planned Reroute
Lunt	Medium	Planned Reroute

Major Delays

Minor Delays

Minor Delays/Reroute

Service Change

Planned Reroute

Planned/Part Closure

Added Service

Bus Stop Relocation

Special Note

Bus Stop Note

Normal Service

Disrupted Services

56

Persistent Bus Routes

51

Out of 127

Persistent Train Route

4

Out of 8

Persistent Systemwide Route

1

Out of 3

Tracking 138 routes (127 Bus · 8 Rail · 3 Systemwide)

Persistence by Severity Level

1-1920-3940-59

1

38

12

Bus Route Alert Summary

ID	Route Name	Severity Level	Persistent Status Category	Route Status
10	Obama Presidential Center/Museum of Science & Industry Express	20-39	Disruption	Planned Reroute
11	Lincoln	20-39	Disruption	Bus Stop Relocation
115	Pullman/115th	20-39	Disruption	Planned Reroute
12	Roosevelt	1-19	Disruption	Bus Stop Note
120	Ogilvie/Streeterville Express	1-19	Disruption	Bus Stop Note
121	Union/Streeterville Express	1-19	Disruption	Bus Stop Note
124	Navy Pier	20-39	Disruption	Planned Reroute
125	Water Tower Express	20-39	Disruption	Planned Reroute
126	Jackson	20-39	Disruption	Planned Reroute
146	Inner Lake Shore/Michigan Express	20-39	Disruption	Planned Reroute
148	Clarendon/Michigan Express	20-39	Disruption	Planned Reroute
15	Jeffery Local	20-39	Disruption	Planned Reroute
157	Streeterville/Taylor	20-39	Disruption	Planned Reroute
172	U. of Chicago/Kenwood	20-39	Disruption	Planned Reroute
18	16th-18th	40-59	Delay	Minor Delays / Reroute
192	U. of Chicago Hospitals Express	20-39	Disruption	Planned Reroute
2	Hyde Park Express	20-39	Disruption	Planned Reroute
20	Madison	1-19	Disruption	Bus Stop Note
22	Clark	1-19	Disruption	Bus Stop Note
24	Wentworth	1-19	Disruption	Bus Stop Note
29	State	20-39	Disruption	Planned Reroute
3	King Drive	1-19	Disruption	Added Service
35	31st/35th	1-19	Disruption	Added Service

Bus

Train & Systemwide

Major Delays

Minor Delays

Minor Delays/Reroute

Service Change

Planned Reroute

Planned /Part Closure

Added Service

Bus Stop Relocation

Special Note

Bus Stop Note

Normal Service

System Health

60.7%▲

System Delay Rate

1.9%▼

Disruption Rate

39.3%▼

Disrupted Routes

23.2%▼

Date

All

Weekday

All

TrainRouteName

All

BusRouteName

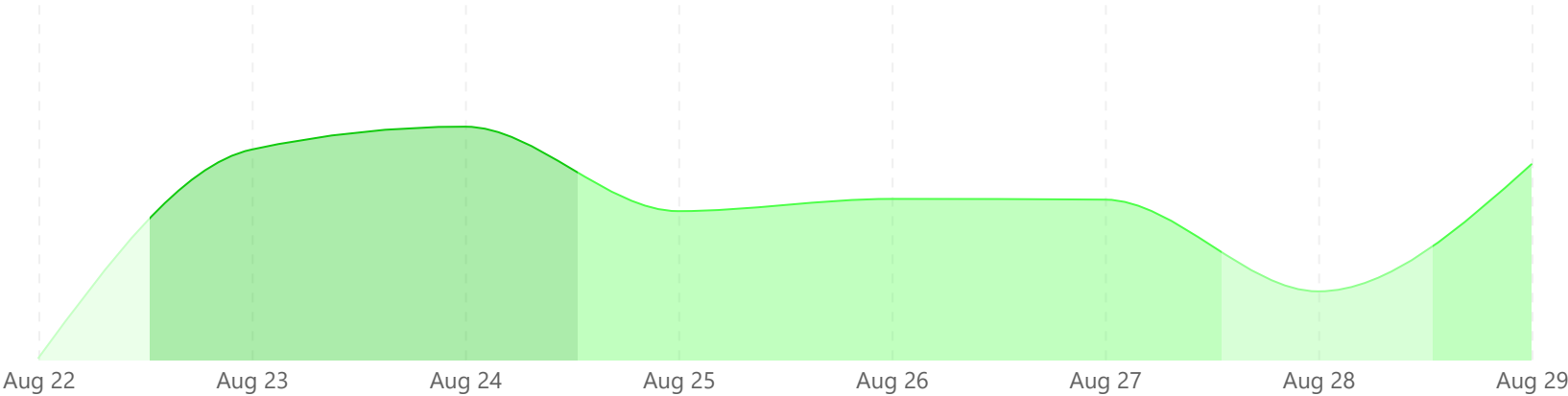
All

Normal Service Count by Total Alerts

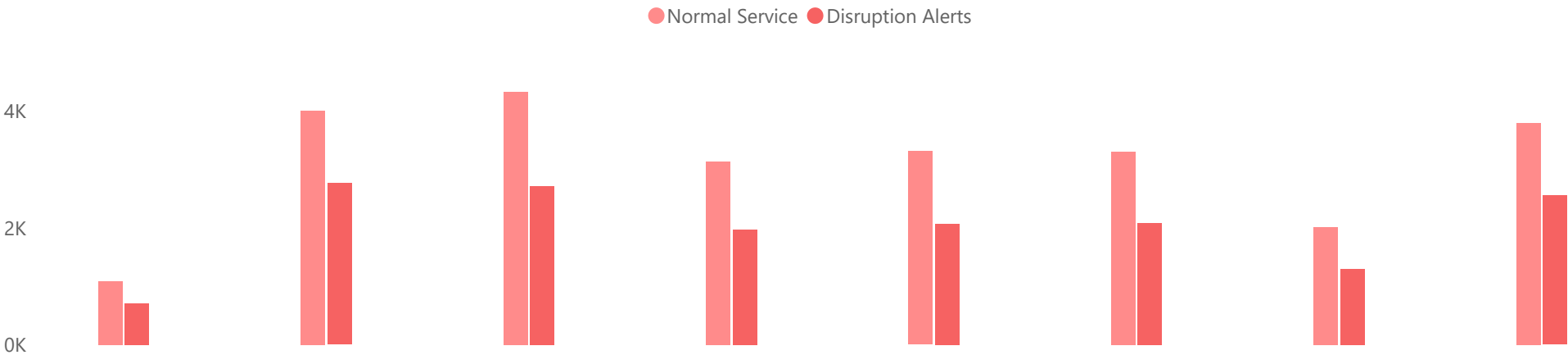
System Health

Delay

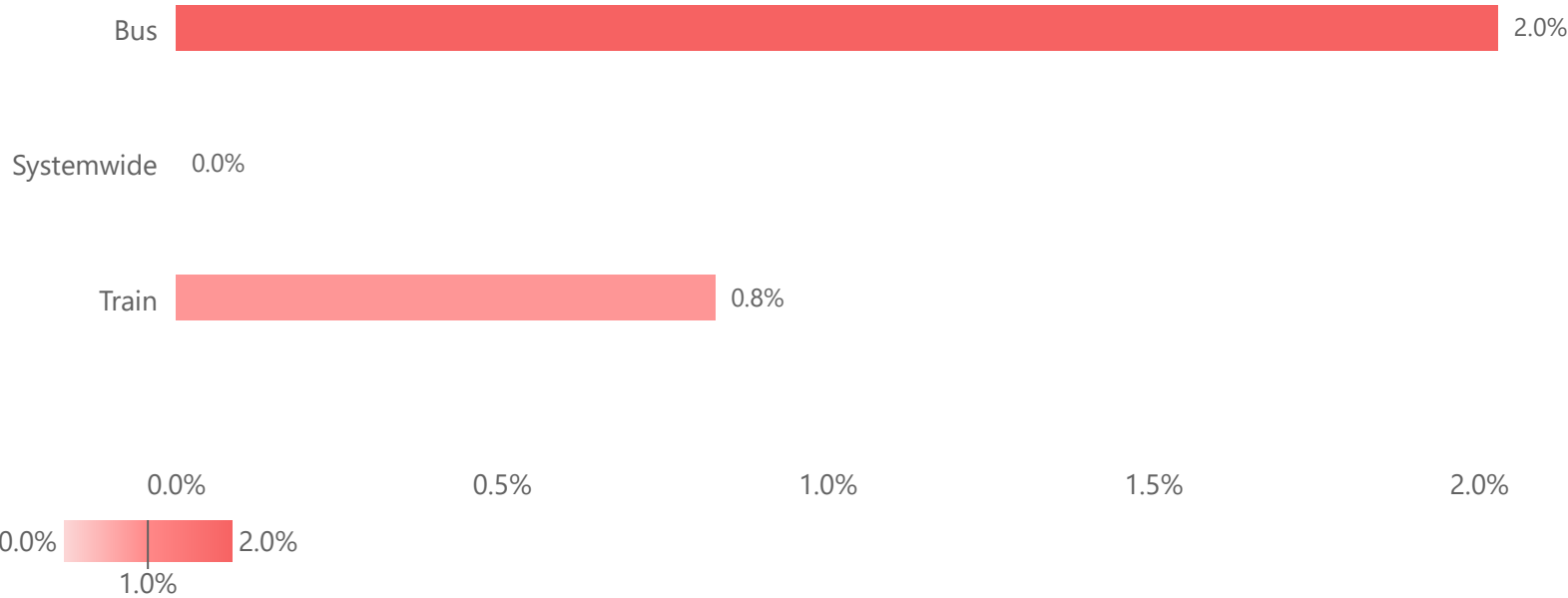
Disruption



Normal Service & Disruption Alerts Per Day



Delay Rate by Service



Alert Summary by Status Category

Alert Status	Category	Disruption Alerts	%Disruptions
Planned Reroute	Planned Change	10,279	25.0%
Added Service	Information	2,316	5.6%
Bus Stop Note	Information	1,699	4.1%
Service Change	Planned Change	743	1.8%
Bus Stop Relocation	Planned Change	390	0.9%
Minor Delays / Reroute	Delays	298	0.7%
Total	Delays	16,158	39.3%

Delay Rate

2.0%▼

Affected Routes

29

Peak Daily Hour

5

Highest Day Period

Afternoon

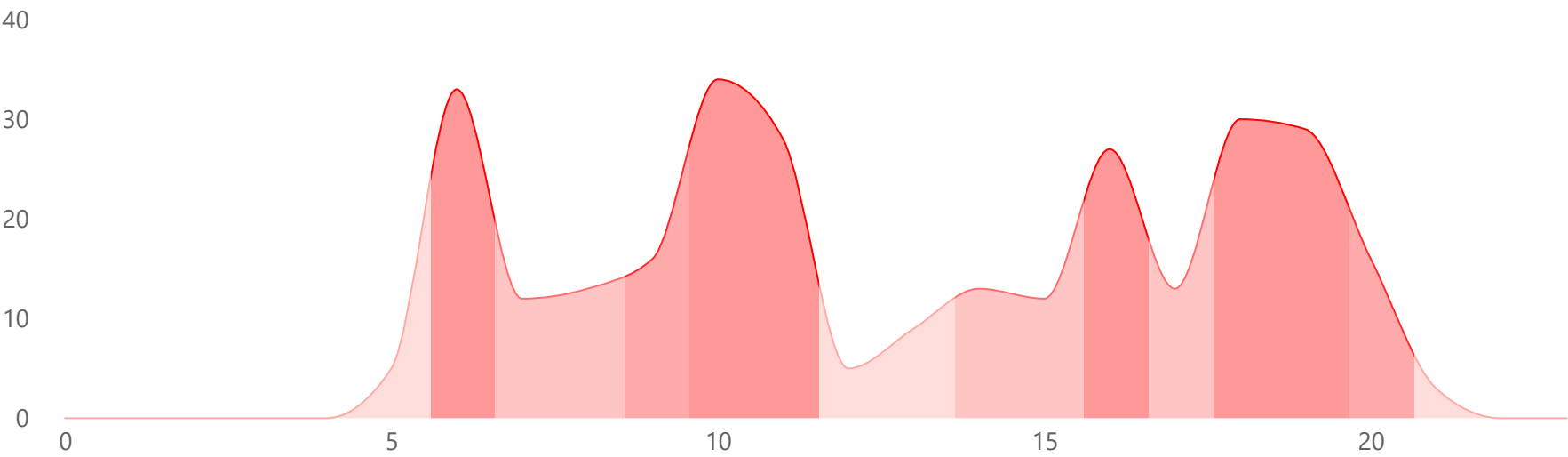
Highest Weekday

Saturday

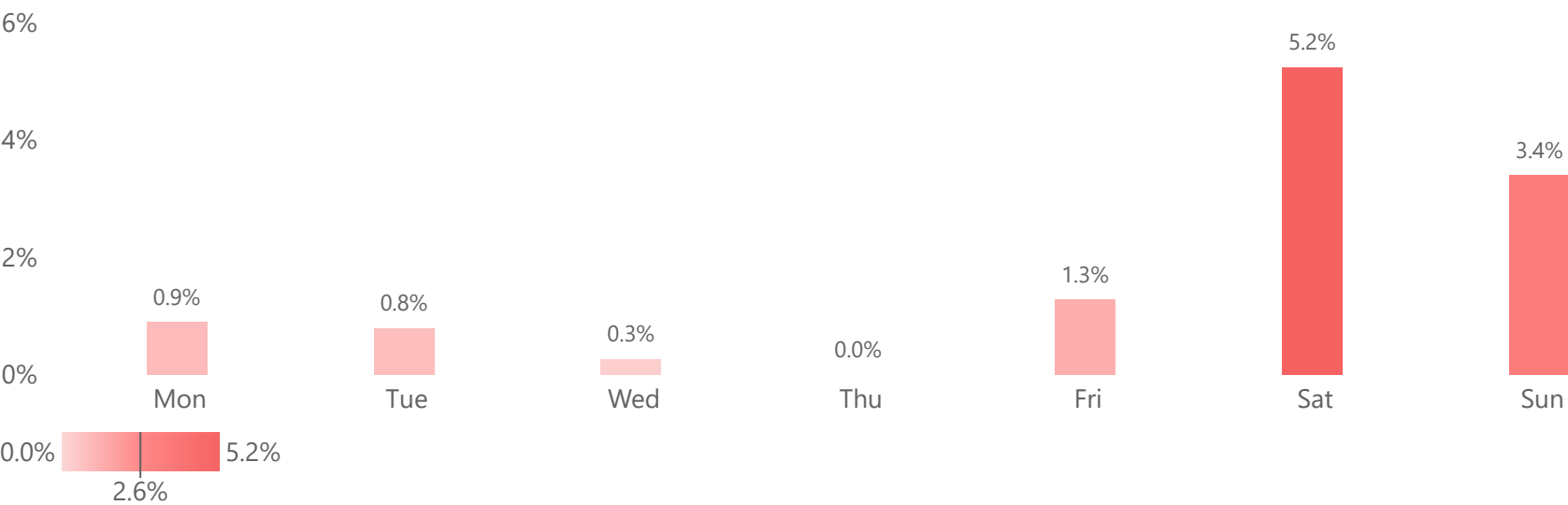
Bus

Train

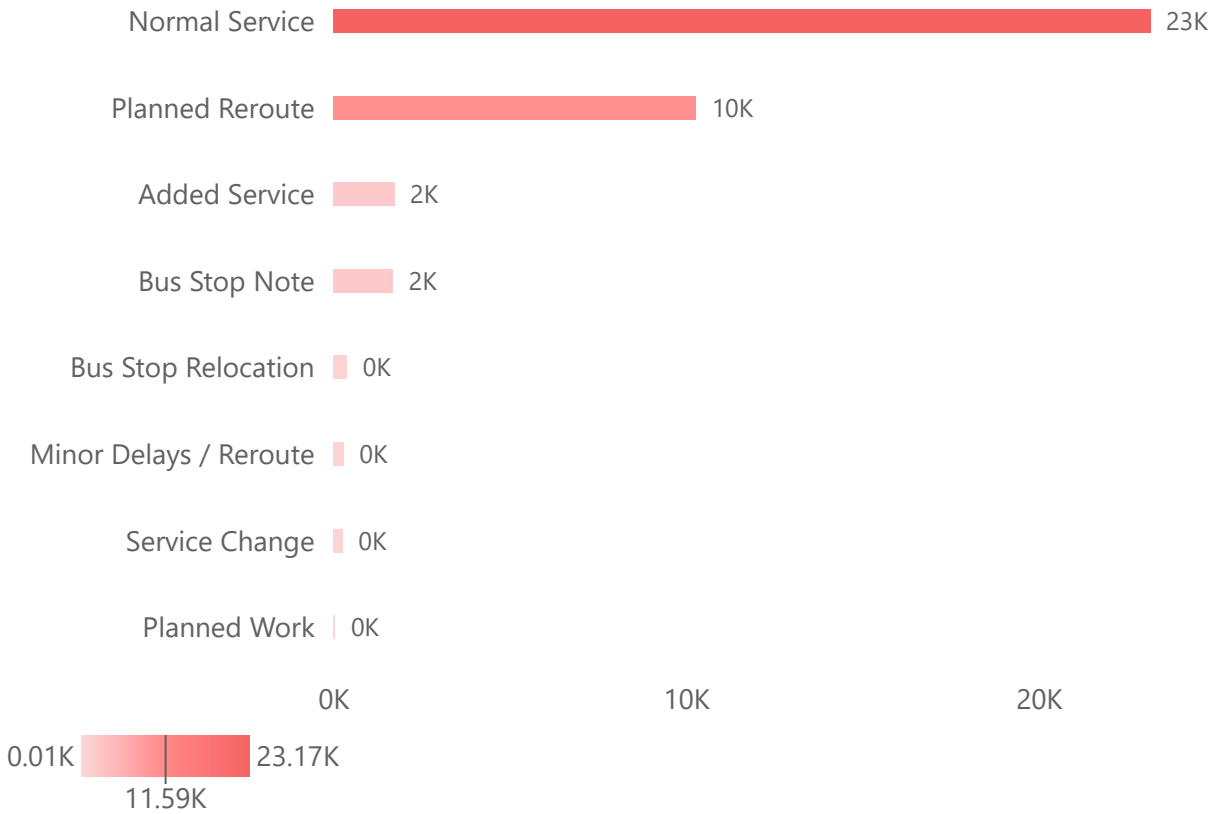
Total Delays by Hour



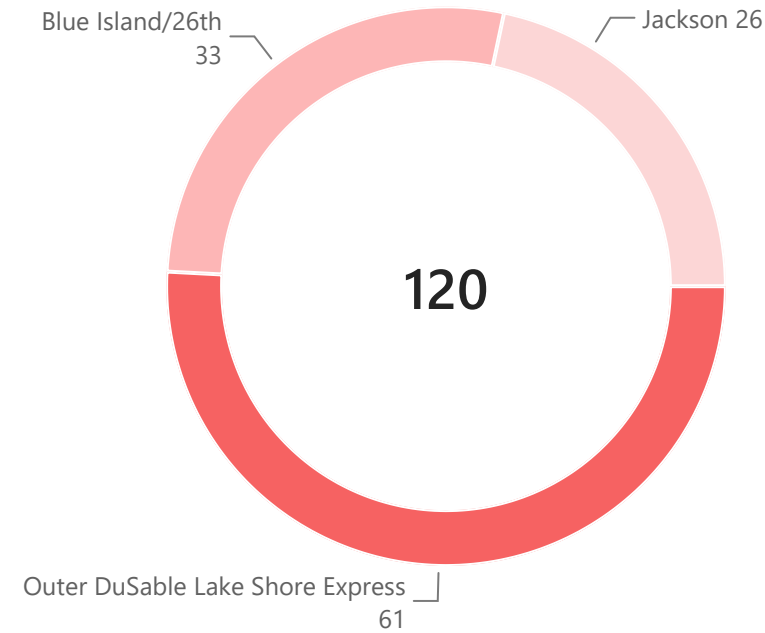
Delay Rate by Weekday



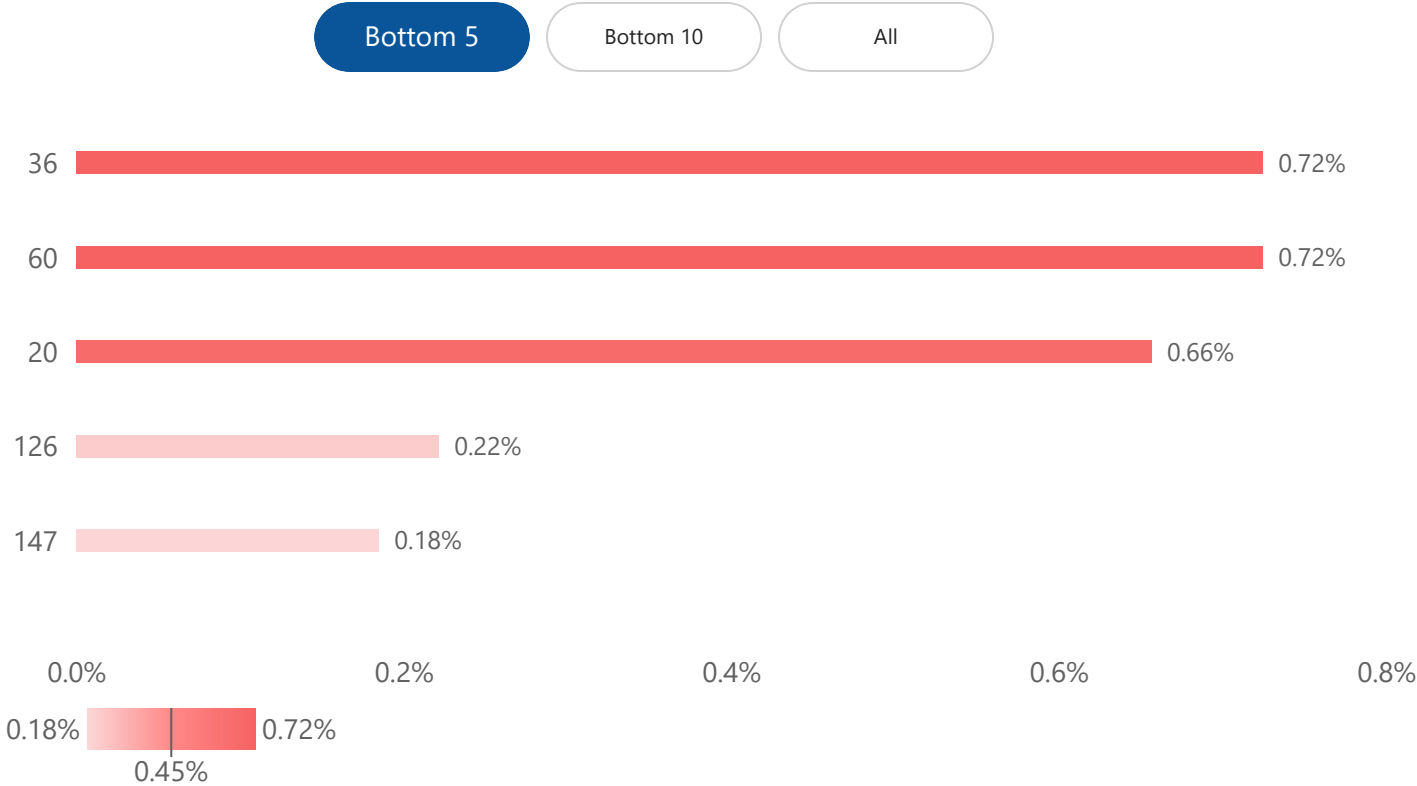
Status Distribution by Alerts



Bottom 3 Bus Lines by Delays



Bus Route Disruption Rate



Recommendations

Introduction

Status (Live)

Persistence (Live)

Overview (Historical)

Performance (Historical)

Recommendations

22-Aug-26 to 29-Aug-26

Disruption Rate

39.3%▼

System Delay Rate

1.9%▼

Bus Delay Rate

2.0%▼

Train Delay Rate

0.8%▼

Disrupted Routes

23.2%▼

Overall

- CTA's system health is 60.7%
- Disruption rate is 39.3% & delay rate is 1.9%.

- Disruption rates are not delays, they should be tracked and communicated separately depending on their severity level.

- Use clear status categories in rider-facing displays.
- Improve rider understanding of service impact and help operations interpret each status more clearly.

Bus Delay Rate

- Bus delay rate is 2.0%
- Peak delays: 5-6 AM & 6-7 PM
- Highest day period: Afternoon | Highest day: Saturday
- Top 3 routes: Blue Island/26th, Jackson 26, Outer DuSable Shore Express

- Delays are concentrated in specific periods and routes, indicating areas for targeted investigation.

- Investigate traffic, running time, bunching, and operator/fleet availability.
- Prioritize affected routes and periods in CTA's running-time review for targeted improvements.

Train Delay Rate

- Train delay rate is 0.8%.
- Peak delays is 10 AM, 4 PM & 7 PM.
- Highest day period: Afternoon | Highest day: Wednesday
- Top 3 train lines: Blue, Green & Pink

- Delays are concentrated on specific lines and time periods, indicating areas for targeted investigation.

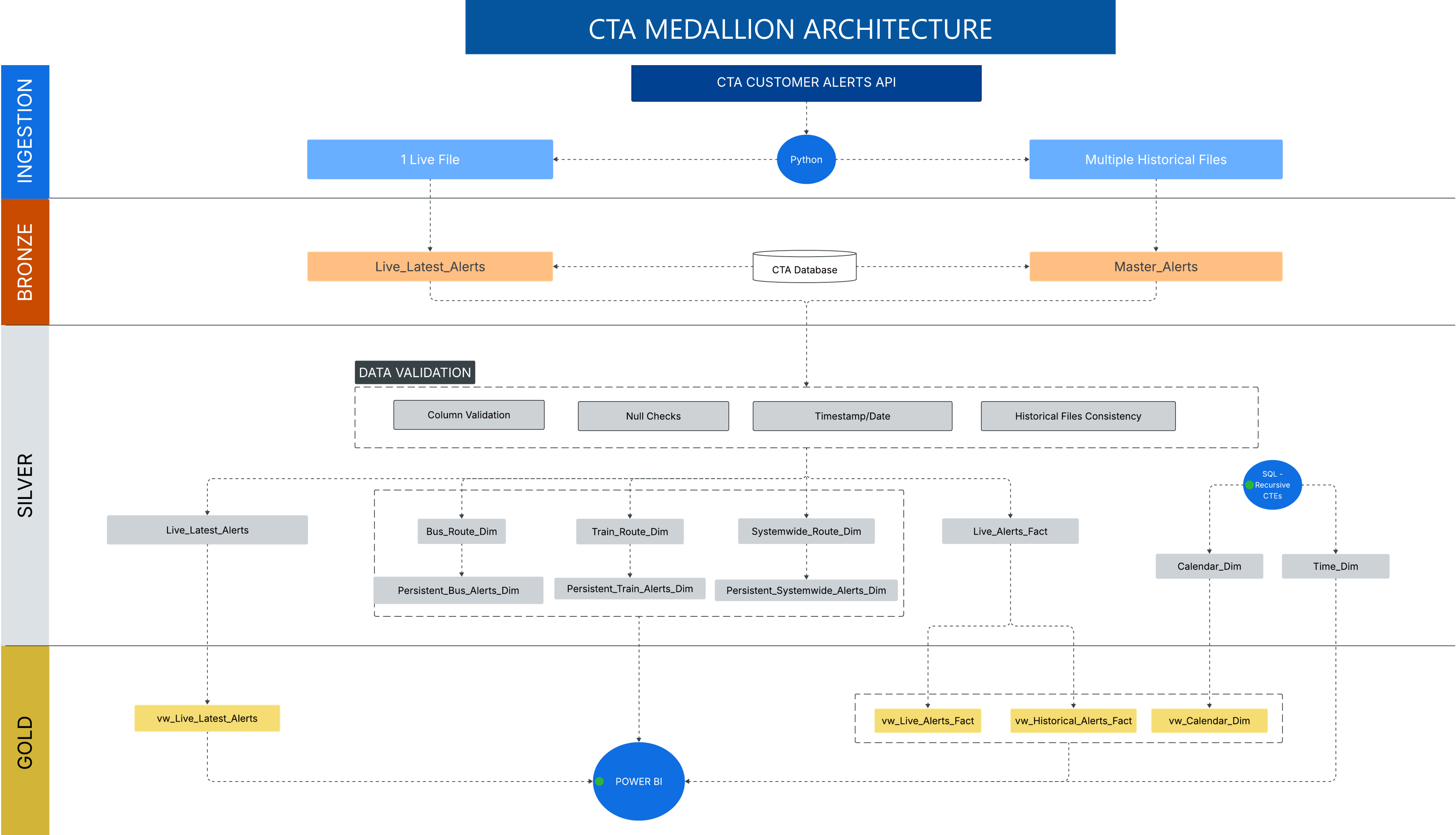
- Investigate slow zones, running time, and other operational factors.
- Compare affected lines and peak periods with CTA's slow-zone elimination work to reduce delays.

Route Statuses

- High disruption-alert volumes on certain routes may indicate recurring service issues.

- Recurring alerts may have operational or external causes.

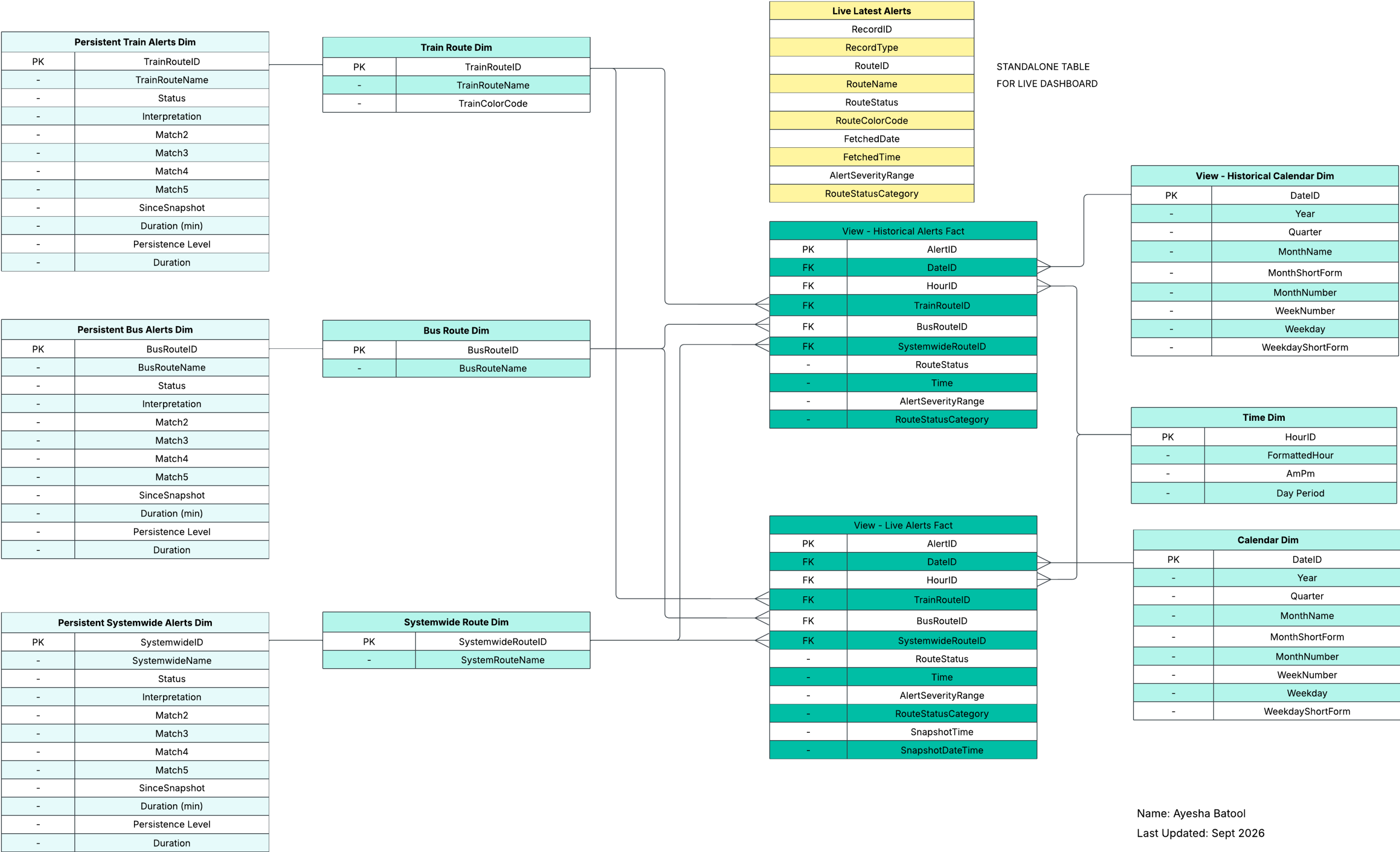
- Prioritize high-alert routes for root-cause analysis.
- Focus on reducing recurring service disruptions, not the number of alerts reported.



ENTITY RELATIONSHIP DIAGRAM (ERD)

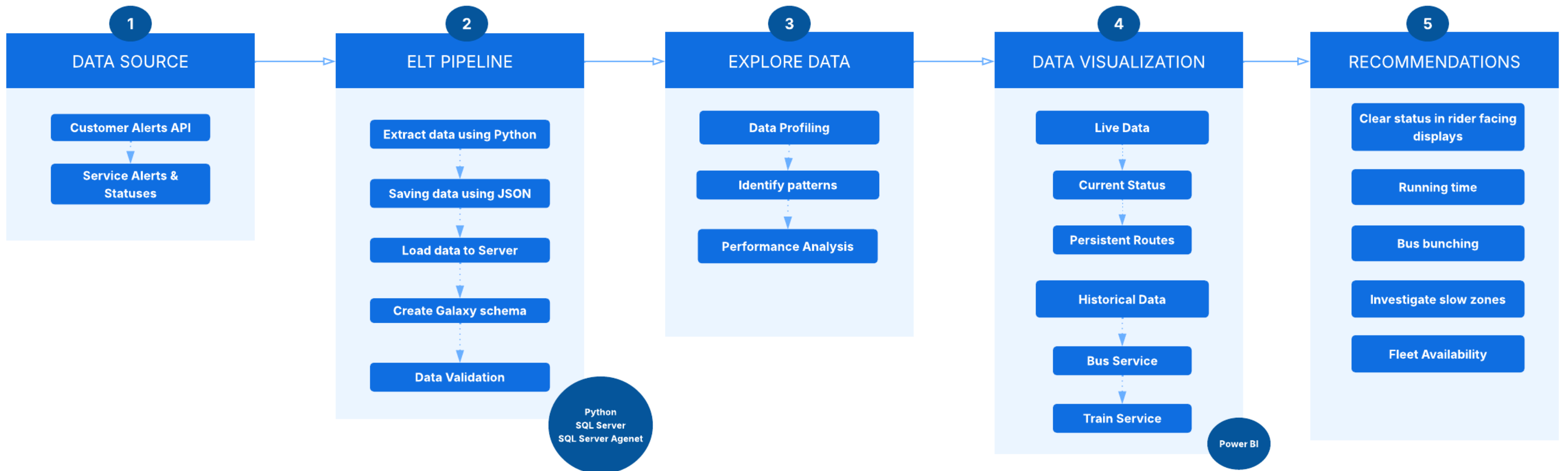
Galaxy Schema

Chicago Transit Authority (CTA) Service Reliability Analysis

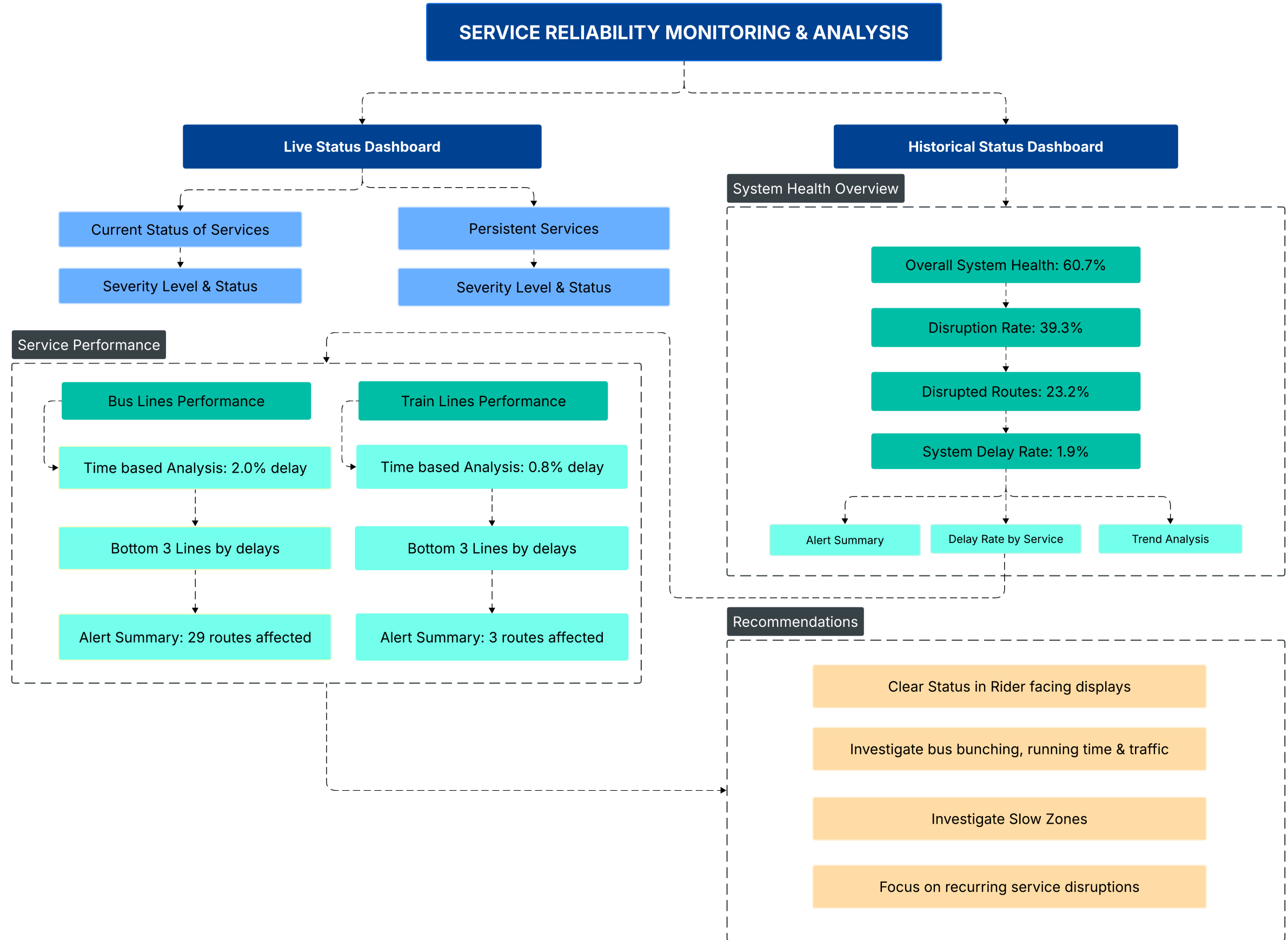


PROJECT WORKFLOW

Chicago Transit Authority (CTA) Monitoring & Reliability



ANALYTICAL WORKFLOW



DATA DICTIONARY

Column Name	Data Type	Description	Key	Table Name
BusRouteID	Text	Unique identifier for each bus route	PK	BusRoute
BusRouteName	Text	Name of the bus route	-	BusRoute
DateID	Date	Unique identified used to link records to the Calendar table	PK	Calendar Dim
MonthName	Text	Full name of the month	-	Calendar Dim
MonthNumber	Whole Number	Numeric representation of the month (1-12)	-	Calendar Dim
MonthShortForm	Text	3 letter abbreviation of the month, used for concise reporting & visualizations	-	Calendar Dim
Quarter	Whole Number	Quarter of the year (Q1-Q4)	-	Calendar Dim
Weekday	Text	Name of the day of the week	-	Calendar Dim
WeekdayShortForm	Text	Abbrevated name of the day of the week, used for concise reporting & visualizations	-	Calendar Dim
WeekNumber	Whole Number	Week number of the year	-	Calendar Dim
Year	Whole Number	Calendar year of the date	-	Calendar Dim
AlertID	Whole Number	Unique identified for each alert record	PK	Historical Alerts Fact
AlertSeverityRange	Text	Severity level associated with the alert eg 0, 1-19, 20-39	-	Historical Alerts Fact
BusRouteID	Text	Links each alert to the BusRoute table	FK	Historical Alerts Fact
DateID	Date	Links each alert to the Calendar table	FK	Historical Alerts Fact
HourID	Whole Number	Links each alert to the Time table	FK	Historical Alerts Fact
RouteStatus	Text	Status of the route at the time of the alert	-	Historical Alerts Fact
RouteStatusCategory	Text	Category grouping assigned to the route status	-	Historical Alerts Fact
SystemwideRouteID	Text	Links each alert to the SystemwideRoute table	FK	Historical Alerts Fact
Time	Time	Timestamp indicating when the alert occurred	-	Historical Alerts Fact
TrainRouteID	Text	Links each alert to the TrainRoute table	FK	Historical Alerts Fact